

Risk Arbitrage Opportunities for Stock Index Options*

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Abstract

To analyze the economic significance of pricing errors of stock index options, a system of linear inequalities is developed which completely characterizes all risk arbitrage opportunities which arise if a well-behaved pricing kernel does not exist. The Stochastic Arbitrage system can account for market imperfections in the form of transactions costs and general portfolio restrictions. An active trading strategy based on the Stochastic Arbitrage system for front-month S&P500 stock index options yields significant abnormal returns out of sample, for small-scale portfolios. However, outperformance seems elusive if the strategy is scaled up and market depth is taken into account.

Key words: Options Pricing, Risk Arbitrage; Options Trading; Stochastic Dominance; Linear Programming

JEL codes: C61; C62; D53; D81; G11; G13

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1 Introduction

A large body of literature deals with the pricing and trading of financial options. Empirical problems encountered with standard models based on complete and perfect markets have motivated research which accounts for incompleteness and frictions. These empirical problems include the option pricing kernel puzzle (Ait-Sahalia and Lo (2000) and Jackwerth (2000)). Stock index option prices imply a U-shaped pricing kernel, after the stock market crash of 1987, which challenges equilibrium asset pricing models based on a risk-averse representative investor.

Constantinides, Jackwerth and Perrakis (2009; CJP) forward an elegant and tractable system of linear inequalities for testing the existence of a well-behaved kernel for the pricing of a cross-section of concurrent, European-style index options in the presence of transactions costs. This pricing kernel system is motivated by the first-order optimality conditions for an investor who holds the underlying index. These optimality conditions ensure that the index is not dominated by any feasible alternative portfolio by the second-degree stochastic dominance (SSD) criterion (Hadar and Russell (1969); Hanoch and Levy (1969); Rothschild and Stiglitz (1970)).

The CJP system is a natural generalization of SD pricing bounds for individual option series in imperfect markets (Constantinides and Perrakis (2002, 2007)); the latter in turn generalize pricing bounds in frictionless markets (Perrakis and Ryan (1984); Levy (1985); Perrakis (1986); Ritchken (1985); Ritchken and Kuo (1988)).

CJP document widespread violations of the system for S&P500 stock index (SPX) options. These violations suggest that the option pricing kernel puzzle cannot be explained by accounting for incompleteness and frictions.

Motivated by these results, Constantinides, Czerwonko, Jackwerth and Perrakis (2011) and Constantinides, Czerwonko and Perrakis (2019) develop option trading strategies. The former study considers strategies based on violations of pricing bounds for individual option

series. The latter study builds portfolios which obey a sufficient condition for dominance of the market index by SSD. The sufficient condition is based on the ‘single-crossing rule’ for the cumulative distribution functions (CDFs) of the optimal portfolio and the market index, and the requirement that the optimal portfolio returns and index returns are co-monotonic.

The above studies form the quasi-totality of the empirical option studies that take into account the truly observed stock index option prices. All other empirical option market studies use models tailored to the frictionless market and modify the data to make it fit to their models. For instance, they assume put-call parity, which may not hold in the data, and use only out-of-the money options to represent the entire cross section.

The present study contributes to this literature by developing a complement to the CJP system which completely characterizes all portfolios which dominate the index by SSD. Compared with Constantinides, Czerwonko, Jackwerth and Perrakis (2011), the analysis is based on joint restrictions for a cross-section of option series instead of bounds for individual series. Compared with Constantinides, Czerwonko and Perrakis (2019), the analysis enlarges the set of risk arbitrage opportunities by using necessary and sufficient conditions which require neither the single-crossing rule nor the co-monotonicity condition.

The relevant risk arbitrage opportunities are formally defined: a Stochastic Arbitrage (SA) opportunity arises when it is possible to engineer a self-financing overlay for the stock index which exhibits SSD, accounting for transactions costs and relevant portfolio constraints.

A linear system is developed to identify all SA opportunities if the option market is not in equilibrium. The SA system is a logical complement to the CJP system: an SA opportunity exists if and only if a well-behaved pricing kernel does not exist. This result extends the analysis by Dybvig (1988) and Beare (2011) from FSD and frictionless markets to SSD and imperfect markets.

The analysis is also reminiscent of the duality between the mean-standard deviation frontier for the pricing kernel and the mean-variance portfolio optimization problem (Hansen

and Jagannathan (1991)). The use of SD instead of mean-variance dominance in the present study is motivated by the inherent asymmetry of option returns.

The logical complementarity between pricing kernels and SA opportunities ties together the aforementioned studies of pricing and trading of stock index options using SSD. The infeasibility of the CJP pricing kernel system lies at the root of the trading profits in Constantinides, Czerwonko, Jackwerth and Perrakis (2011) and Constantinides, Czerwonko and Perrakis (2019), something which these studies seem to have overlooked.

The SA system goes beyond the ‘simple’ application of linear duality theory to the CJP system. While the SA system describes all feasible and dominant solutions, the dual alternative to the CJP system gives only a necessary but not sufficient condition. In addition, the SA system has more discriminatory power than the analysis of Constantinides, Czerwonko and Perrakis (2019), because the optimal solution generally violates the single-crossing rule and the co-monotonicity condition.

The SA system facilitates portfolio restrictions which are not binding for a passive position in the index but which may become binding for active portfolios with open positions in options, for instance, restrictions based on market depth in the limit order book. Such restrictions can enhance the analysis of the economic significance of pricing errors of stock index options.

The implementation of the SA system can draw on an existing literature on portfolio choice with SSD constraints. The numerical optimization has been studied previously by Dentcheva and Ruszczyński (2003), Kuosmanen (2004), Roman, Darby-Dowman and Mitra (2006), Kopa and Post (2015) and Longarela (2016). To obtain a discrete estimator of the conditional return distribution, lattice models such as the binomial tree model by Cox, Ross and Rubinstein (1979) can be used with model parameters calibrated to the prevailing market conditions, following CJP and the aforementioned empirical studies of option trading strategies.

Instead of using the proposed nonparametric approach based on SSD, optimization of

option combinations can be based on parametric specifications of investor risk preferences. For example, Faias and Santa-Clara (2017) develop a method to optimize a portfolio of European-style options for a given power utility function. An advantage of the SSD approach is a higher robustness to specification error and sampling variation.

The flip side of the SSD approach is a possible loss of discriminatory power. However, the application to concurrent index options benefits from the natural single-factor payoff structure which avoids the curse of dimensions which plagues applications of SSD to cross-sections of primary securities.

In an empirical application to front-month SPX options, an active trading strategy based on the solutions to the SA system is evaluated. The investment performance depends critically on the options' time to maturity and, furthermore, the assumed portfolio size. SA opportunities are larger for front-month options than for back-month options. Large-scale implementation does not outperform a passive position in the SPX, if market depth restrictions from the limit order book are taken into account. Consequently, the significant pricing errors found in previous research do not suffice to prove the existence of large-scale risk arbitrage opportunities.

2 Theory

2.1 Preliminaries

The investment universe includes M securities: a Treasury bill, a stock index and $(M-2)$ concurrent series of European options on the index.

The value of a given security at option expiration is denoted by x_i , $i = 1, \dots, M$. The bill matures on the options expiration date and has principal x_1 . The stock index pays cash dividends $d \geq 0$; the cum-dividend price is $x_2 = y$ and the ex-dividend price is $(y - d)$. Options have strikes k_i and contingent payoffs $x_i = \max(y - d - k_i, 0)$ for calls or $x_i =$

$\max(k_i - y + d, 0)$ for puts, $i = 3, \dots, M$.

Securities can be bought at a net purchase price (ask price plus trading fees) of $a_i \geq 0$ and sold or written at a net sales price (bid price minus trading fees) of b_i , $a_i \geq b_i \geq 0$. The empirical analysis in this study assumes a zero bid-ask spread for the bill, so that $a_1 = b_1 = R^{-1}x_1$, where R is the gross risk-free rate during the life of the options. Similarly, the index is assumed to have a zero bid-ask spread, so that $a_2 = b_2 = y_0$, where y_0 is the current value of the index.

No costs are involved for principal payments and cash settlements at the expiration date. Whether the index is traded at the expiration date depends on the assumed reinvestment and refinancing possibilities. For simplicity, it is assumed here that the unadjusted stocks position at expiration is carried to the next period, so that the end-of-period costs are zero. It is straightforward to adjust the cost structure in relevant cases, after deriving the SA system for the present specification.

A tractable discrete representation of the joint distribution of $\mathbf{x} = (x_1 \cdots x_M)'$ is used, to allow for the use of discrete mathematics and numerical optimization. The univariate payoff distribution of the index is represented by a multinomial, $\mathcal{F}$, with atoms at y_j , $y_1 < \cdots < y_N$, and physical probabilities $p_j > 0$, $j = 1, \dots, N$.

2.2 Pricing kernel system

The CJP system requires the existence of a non-negative and anti-monotonic pricing kernel, $m(y)$: $m(y_1) \geq \cdots \geq m(y_N) \geq 0$. To facilitate compact presentation, the pricing kernel system in this study is presented both in summation notation and in matrix notation.

Using m_j , $j = 1, \dots, N$, for model variables which represent the values of the kernel, the formulation in summation notation (first introduced in Ritchken (1985)) follows:

$$\begin{aligned}
& \sum_{j=1}^N p_j x_{ij} m_j \leq a_i, \quad i = 1, \dots, M; \\
& - \sum_{j=1}^N p_j x_{ij} m_j \leq -b_i, \quad i = 1, \dots, M; \\
& m_j - m_{j+1} \geq 0, \quad j = 1, \dots, N-1; \\
& m_j \geq 0, \quad j = 1, \dots, N.
\end{aligned} \tag{1}$$

The equivalent formulation in matrix notation uses $\mathbf{P} := (\mathbf{1}_M \mathbf{p})$, $\mathbf{p} := (p_1 \cdots p_N)$, $\mathbf{X} := (\mathbf{x}_1 \cdots \mathbf{x}_N)$, $\mathbf{x}_j := (x_{1j} \cdots x_{Mj})'$, $\boldsymbol{\mu} := (m_1 \cdots m_N)'$, $\mathbf{a} := (a_1 \cdots a_M)'$, $\mathbf{b} := (b_1 \cdots b_M)'$, $\mathbf{D}$ for an upper triangular difference matrix and the entrywise product ‘ $\circ$ ’:

$$\begin{aligned}
& (\mathbf{P} \circ \mathbf{X}) \boldsymbol{\mu} \leq \mathbf{a}; \\
& -(\mathbf{P} \circ \mathbf{X}) \boldsymbol{\mu} \leq -\mathbf{b}; \\
& \mathbf{D} \boldsymbol{\mu} \geq \mathbf{0}_N; \\
& \boldsymbol{\mu} \geq \mathbf{0}_N.
\end{aligned} \tag{2}$$

The system is equivalent to Constantinides, Jackwerth and Perrakis (2009, Eq. (1)-(9), p. 1253-1255), modulo the omission of end-of-period costs for trading stocks. It is straightforward to adjust the assumed investment problem after the corresponding SA system is derived.

The system can be interpreted as a set of first-order optimality conditions for a myopic and risk averse investor who is fully invested in the index or a position which is co-monotone to the index. The kernel represents the Intertemporal Marginal Rate of Substitution for this

investor.

The system is relatively small, because the kernel is evaluated only at the atoms of the distribution of the index, y_j , $j = 1, \dots, N$. These points are known in advance and are ranked ($y_1 < \dots < y_N$), which allows for fixing the ranking of the kernel values ($m(y_1) \geq \dots \geq m(y_N)$) and avoids the need to endogenize the ranking.

In practice, violations of the pricing conditions are pervasive for reasonable assumptions about transactions costs. A relevant question seems whether the violations are economically significant. The next section provides a complementary system which can be used to answer this question.

The analysis differs from Constantinides, Czerwonko, Jackwerth and Perrakis (2011) by building on the CJP system rather than the pricing bounds for individual options by Constantinides and Perrakis (2007); it extends Constantinides, Czerwonko and Perrakis (2019) by using necessary and sufficient conditions, rather than sufficient conditions, for SSD. The analysis therefore identifies a larger set of risk arbitrage opportunities than the previous studies.

2.3 Stochastic Arbitrage system

The investor benchmarks against a long position in the stock index and contemplates deviating from the index by constructing a layover portfolio. The decision variables are the numbers of securities bought at the ask price, $\alpha \geq \mathbf{0}_M$, and the numbers of securities sold or written at the bid price, $\beta \geq \mathbf{0}_M$; these numbers need not be natural numbers. The payoff to the layover is given by $x = \mathbf{x}'\alpha - \mathbf{x}'\beta$; adding the layover to the index yields the enhanced portfolio payoff $z = x + y$.

Examples of basic options strategies include the writing of covered calls and the buying of protective puts; more complex strategies combine multiple option types and strikes to build, for example, butterfly spreads and condors.

The feasible combinations for the layover portfolio are defined by a polytope of general

form, $\mathcal{P} := \{(\boldsymbol{\alpha}, \boldsymbol{\beta}) \in \mathbb{R}^M \times \mathbb{R}^M : \mathbf{A}\boldsymbol{\alpha} + \mathbf{B}\boldsymbol{\beta} \leq \mathbf{c}\}$, where $\mathbf{A}$ and $\mathbf{B}$ are the LHS and $\mathbf{c}$ are the RHS coefficients of the portfolio constraints. The solution $(\boldsymbol{\alpha}, \boldsymbol{\beta}) = (\mathbf{0}_M, \mathbf{0}_M)$ is assumed to be feasible and to have no binding constraints, so that $(\epsilon \mathbf{1}_M, \epsilon \mathbf{1}_M) \in \mathcal{P}$ for sufficiently small $\epsilon > 0$.

SSD is used to provide a partial ordering of the feasible portfolios. SSD is formulated here in terms of the lower partial moment function $L(x|t) := \mathbb{E}_{\mathcal{F}}[(t - x)\mathcal{I}(x \leq t)]$ (Bawa (1975); Fishburn (1977); Ritchken (1985)). Due to the use of a discrete distribution, this function is piecewise-linear: $L(x|t) = \sum_{j=1}^N p_j (t - x_j) \mathcal{I}(x_j \leq t)$.

Definition 2.3.1 (SSD): *Payoffs z dominate payoffs y by SSD, or, $z \succeq_{SSD} y$, if $L(z|y_j) \leq L(y|y_j)$ for all $y_j, j = 1, \dots, N$.*

Due to its functional properties, the lower partial moment function needs to be evaluated only at the atoms $y_j, j = 1, \dots, N$.

A well-known equivalent formulation is that SSD occurs if the first alternative achieves a higher expected utility than the second alternative for all nonsatiable and risk averse investors.

Examples of options strategies aimed to create SSD of the index include the covered writing of call options and bull spreads and investing the option premium income in bills or stocks. These strategies sacrifice upside potential to reduce downside risk. Whether the premium income suffices to establish SSD of course depends on the prevailing bid and ask prices. The covered writing of butterfly or condor spreads also reduces downside risk but in exchange for lower returns during sideways markets instead of market upswings.

Definition 2.3.2 (Stochastic Arbitrage): *A Stochastic Arbitrage opportunity arises if a self-financing layover $x = \mathbf{x}'\boldsymbol{\alpha} - \mathbf{x}'\boldsymbol{\beta}$ can be formed such that net costs are negative ($\mathbf{a}'\boldsymbol{\alpha} - \mathbf{b}'\boldsymbol{\beta} < 0$) and the payoffs of the enhanced portfolio $z = x + y$ dominate the payoffs of*

the underlying index y by SSD:

$$\begin{aligned} \mathbf{x}'\boldsymbol{\alpha} - \mathbf{x}'\boldsymbol{\beta} + y &\succeq_{SSD} y; \\ \mathbf{a}'\boldsymbol{\alpha} - \mathbf{b}'\boldsymbol{\beta} &< 0; \\ (\boldsymbol{\alpha}, \boldsymbol{\beta}) &\in \mathcal{P}. \end{aligned} \tag{3}$$

In contrast to the CJP system (2), the SA system (3) is not linear due to the binary variables $\mathcal{I}(x_j \leq t)$, $j = 1, \dots, M$, in the lower partial moment function. However, Section 3.2 provides an equivalent linear system using a known linearization of the lower partial moment. The linearized system is larger and computationally more demanding than the CJP system, but it also generates additional information.

The two systems can be shown to be logical complements:

Theorem 2.3.3: *A Stochastic Arbitrage opportunity (or solution to (3)) exists if and only if no well-behaved pricing kernel (or solution to (2)) exists.*

The two systems thus lead to identical conclusions regarding option market efficiency. However, in contrast to the CJP system (2), the SA system (3) defines all feasible SA opportunities, which is useful to analyze the economic significance of pricing errors and engineer option positions in practice.

The logical complementarity between pricing kernels and SA opportunities seems a missing link in the literature about pricing and trading stock index options using SSD. The empirical violations of the pricing kernel system documented by CJP lies at the root of the trading profits documented by Constantinides, Czerwonko, Jackwerth and Perrakis (2011) and Constantinides, Czerwonko and Perrakis (2019).

The SA system (3) is not simply the dual alternative to the CJP system (2). The proof to Theorem 2.3.3 in the appendix derives the dual alternative using Farkas' Lemma; see Eq. (7). The dual alternative gives a necessary but not sufficient condition for dominance by

the solution portfolio, similar to the dual formulation of nonparametric portfolio efficiency tests in Post (2003, Thm 2). Among other problems, the dual alternative does not bound the option positions, allowing for unbounded short positions which generally introduce SSD violations and insolvency risk.

Essentially, the CJP system does not account for dominance and feasibility restrictions which are not binding for the index. Since these restrictions have a zero shadow price, they do not affect the optimality classification of the index and they can be ignored for testing the system of pricing restrictions. However, these restrictions may become binding for enhanced portfolios and should therefore be taken into consideration when analyzing economic significance of pricing errors if a well-behaved kernel does not exist.

The SA system (3) builds on the exact definition of SSD. By contrast, Constantinides, Czerwonko and Perrakis (2019) use a sufficient but not necessary condition for SSD. Their analysis allows for only a single crossing of the CDFs $F_z(t) := \mathbb{E}_{\mathcal{F}}[\mathcal{I}(z \leq t)]$ and $F_y(t) := \mathbb{E}_{\mathcal{F}}[\mathcal{I}(y \leq t)]$. The following corollary to Definition 2.3.1 demonstrates that this requirement is generally not harmless:

Corollary 2.3.4 (Multiple-Crossing Rule): *If $z \succeq_{SSD} y$, and, furthermore, $L(z|y_2) < L(y|y_2)$, $L(z|y_j) = L(y|y_j)$, for some $j = 3, \dots, N-1$, and $L(z|y_N) < L(y|y_N)$, then $F_z(t)$ and $F_y(t)$ cross at least twice.*

The optimal portfolio and market index in the empirical application obey the stated premises and, consequently, the two CDFs cross at least twice in every cross-section of options.

The SA system furthermore avoids the need for the condition by Constantinides, Czerwonko and Perrakis (2019, Footnote 7) that optimal portfolio returns and market index returns are co-monotonic. To ensure this condition, the empirical analysis by Constantinides, Czerwonko and Perrakis (2019, Section 4)) limits the total options position to one option

per one unit of index.

Further research could extended the SA system to multi-period investment problems, to complement the system of multi-period pricing restrictions in Appendix A of Constantinides, Jackwerth and Perrakis (2009, p. 1272-1273). Naturally, there is no need to consider more than one period if SA exist under the current definition for buy-and-hold strategies.

3 Methodology

3.1 Probability estimation

Most previous applications of SD in asset pricing and portfolio analysis use nonparametric estimators for the probability vector $\mathbf{p}$. The simplest and most common estimator is the empirical probability mass function of a sample of historical observations. In the presence of side information or dynamic patterns, statistical efficiency can be enhanced using the implied probabilities of Generalized Empirical Likelihood (Qin and Lawless (1994)). Almeida and Garcia (2012, 2017), Julliard and Ghosh (2012), Ghosh, Julliard and Taylor (2016), Post and Poti (2017) and Post, Karabati and Arvanitis (2018) have explored this route.

In the option pricing literature, however, lattice models are a more common approach to obtain discrete estimators. Following CJP and the related empirical studies of option trading strategies, the present study uses a conditional multinomial distribution. The binomial tree model by Cox, Ross and Rubinstein (1979) is used with 100 steps and parameters calibrated, at every formation date, to the observed levels of the T-bill rate and CBOE Volatility Index (VIX); an assumed market risk premium of 4% per annum (excluding dividends); and assumed volatility risk premium of 2% per annum.

A robustness analysis is performed based on the assumption that the market risk premium is 0% or 8% per annum. It seems plausible that the true risk premium changes with the business cycle and financial market conditions. Hence, the analysis also considers spec-

ifications in which the market risk premium is proportional to the concurrent level of the VIX or the square of the VIX. The latter specification assumes that the aggregate coefficient of Relative Risk Aversion (instead of the market risk premium) is constant over time.

The use of extremely small probabilities in the tails of the multinomial can cause numerical instability in the optimization process. For this reason, the atoms of the distribution are truncated at 10 standard deviations below and above the mean and the probabilities of the remaining atoms are winsorized at a level of 10^{-10} (before re-normalization of the probabilities).

An advantage of this lattice model, compared with the aforementioned empirical probabilities or implied probabilities, is that it does not limit the atoms of the distribution to historical scenarios, and uses a support which is much wider (20 standard deviations) than the empirical support, hence mitigating the 'Peso problem'. Furthermore, by calibrating the model parameters to the prevailing market conditions, a conditional, forward-looking estimator is obtained which is expected to have superior predictive ability.

The benefits of further refinements of the probability estimation depend on the relevant set of portfolio constraints. Investment restrictions tend to mitigate the effects of estimation error and yield more robust investment performance; see, for instance, Jagannathan and Ma (2003) and DeMiguel, Garlappi, Nogales and Uppal (2009). Unreported results show that the results and conclusions for large-scale portfolios (with tight market-depth constraints) are indeed very robust to the estimation method, for example, using the aforementioned empirical probabilities or implied probabilities instead of the lattice model, or reasonable changes of the number of steps, truncation of the atoms and winsorization of the probabilities. By contrast, the results are less robust for small-scale portfolios.

3.2 Numerical optimization

The existence of a feasible solution can be determined using numerical optimization. A literature in OR/MS has developed various linear formulations of the SSD relation for

discrete distributions; see Kopa and Post (2015) for a survey and generalization.

Using $\mathbf{L} := (L(y|y_1) \cdots L(y|y_N))'$ and $\mathbf{y} := (y_1 \dots y_N)$, the following linearization is obtained:

Proposition 3.2.1: *Every SA is a solution to the following system of linear inequalities and every solution is an SA opportunity:*

$$\begin{aligned}
 p\Theta &\leq \mathbf{L}; \\
 -\Theta - \mathbf{X}'\alpha\mathbf{1}'_N + \mathbf{X}'\beta\mathbf{1}'_N &\leq \mathbf{y}'\mathbf{1}'_N - \mathbf{1}_N\mathbf{y}; \\
 \mathbf{a}'\alpha - \mathbf{b}'\beta &< 0; \\
 \mathbf{A}\alpha + \mathbf{B}\beta &\leq \mathbf{c}; \\
 \alpha, \beta &\geq \mathbf{0}_N; \\
 \Theta &\geq \mathbf{0}_{N \times N}.
 \end{aligned} \tag{4}$$

The linear system can be tested using Linear Programming (LP) techniques. Notably, an LP problem can be designed for optimizing an objective function subject to the dominance and feasibility constraints (4). The relevant specification of the objective function depends on the application at hand.

The problem specification in the empirical application maximizes initial premium income $(\mathbf{b}'\beta - \mathbf{a}'\alpha)$ subject to the linear system:

$$\begin{aligned}
& \max \mathbf{b}'\boldsymbol{\beta} - \mathbf{a}'\boldsymbol{\alpha} \\
& \mathbf{p}\boldsymbol{\Theta} \leq \mathbf{L}; \\
& -\boldsymbol{\Theta} - \mathbf{X}'\boldsymbol{\alpha}\mathbf{1}'_N + \mathbf{X}'\boldsymbol{\beta}\mathbf{1}'_N \leq \mathbf{y}'\mathbf{1}'_N - \mathbf{1}_N\mathbf{y}; \\
& \mathbf{A}\boldsymbol{\alpha} + \mathbf{B}\boldsymbol{\beta} \leq \mathbf{c}; \\
& \boldsymbol{\alpha}, \boldsymbol{\beta} \geq \mathbf{0}_N; \\
& \boldsymbol{\Theta} \geq \mathbf{0}_{N \times N}.
\end{aligned} \tag{5}$$

If an SA opportunity exists, then the optimal overlay generates initial income of $\mathbf{b}^*\boldsymbol{\beta} - \mathbf{a}^*\boldsymbol{\alpha} > 0$ at portfolio formation, and, furthermore, the enhanced portfolio payoff at option expiration dominates the index payoff by SSD: $\mathbf{x}'\boldsymbol{\alpha}^* - \mathbf{x}'\boldsymbol{\beta}^* + y \succeq_{SSD} y$. On the expiration day, the total holding period return is computed as the final payoff plus the initial income with accrued interest.

The above problem specification is favorable for the robustness of total holding period return, because current income is not conditional on the index and not affected by sampling error. Current income in effect shifts up the entire return distribution by a constant and thus operates as a slack variable for the dominance constraints.

Robustness may be further enhanced by appropriate relaxations of the SA system. Due to the complexity of the SD relation, sampling error has an asymmetric effect: false dominance classifications are less likely to occur than false non-dominance classifications; see Kroll and Levy (1980), Linton, Maasoumi and Whang (2005) and Post, Karabati and Arvanitis (2018). This pattern suggests that relaxing instead of tightening the inequalities can enhance out-of-sample performance; various alternative relaxations are discussed in the Conclusions.

The single-crossing rule used in Constantinides, Czerwonko and Perrakis (2019) tightens the optimization problem instead of relaxing it. Furthermore, their analysis limits the total

options position to one option per one unit of index, to satisfy the co-monotonicity condition. These restrictions are expected to have an adverse effect on out-of-sample performance in the present study.

Since $N \gg M$ in typical applications, the number of variables and constraints is $\mathcal{O}(N^2)$ and increases at a quadratic rate with the number of grid point for partitioning the support of the underlying distribution. However, the problem is perfectly tractable for hundreds of partition points with standard computer hardware and software. For an ultra-fine partition, high-performance platforms and specialized solver software are recommended.

4 Analysis of SPX Options

4.1 Data

A panel data set is used from CBOE's DataShop containing all available best bid and ask prices and sizes for European-style call and put options on the SPX recorded at 14:45 CT (30 minutes before the close at 15:15 CT) from April 1, 2004, through April 1, 2018. In addition, the analysis uses SPX values (source: Yahoo Finance), monthly dividends on the index (source: Robert Shiller's homepage) and T-bill rates and dollar-denominated LIBOR rates (source: Federal Reserve Bank of St. Louis).

The empirical design resembles that of CJP; 28 days before each expiration date, the SA problem (5) is solved for a cross-section of front-month options. The sample period includes 168 formation and expiration dates.

A maturity of 28 days seems appropriate here, for a number of reasons. Longer maturities seem less relevant, because back-month options are known to have relatively small pricing errors. Furthermore, shortening the maturity entails having no open position in options for a larger part of the month, because the analysis does not include weekly SPX options outside of the expiration week of the monthly options. Despite these reservations, an analysis of 91-day options and monthly 7-day options is included.

One of the factors which determines the scope for financial engineering of option combinations is the the number of different option series which is available at the formation date. Table I contains information about the level and variation of this variable. The median number of series is presented separately for call and put, as well as bid and ask and different time periods, maturities and moneyness categories.

Hundreds of different option series are available on the typical formation date. The number of series is relatively low at the beginning of the sample period, for 91-day options, and for high strike prices. For deep out-of-the money series (calls with very high strikes and puts with very low strikes), bid volume is often zero, even if an ask price is quoted.

[Insert Table I about here.]

Data filters based on liquidity and moneyness have a minimal effect in this study, because bid-ask spreads and market depth are explicitly taken into account in the portfolio constraints; see Section 4.2. An exception are options with strikes which lie outside the support of the estimated probability distribution. The valuation of these options is naturally very sensitive to the Peso problem. These extreme options play no material role for large-scale portfolios, because their market depth is limited. Nevertheless, these options are excluded from the analysis to avoid possible distortion of the profitability of small-scale portfolios.

Apart from the number of option series, the quoted volume per option series at the formation date plays an important role in the analysis of large portfolios. Table II contains information about the level and variation of this variable, using the same format as Table I. The difference between bid and ask quote sizes for an option is the market makers' unhedged portion. The differences in quote sizes between calls and puts for the same degree of moneyness points to different markets for calls and puts.

The quote size is generally limited to tens or hundreds of contracts. Not surprisingly,

volume increases over time. Volume also tends to increase with maturity, but the highest values are found for the ask quotes of 7-day out-of-the-money calls and puts. The low median quote size for every type of quote casts doubt on the scalability of investment returns of small-scale option layovers.

[Insert Table II about here.]

4.2 Portfolio constraints

The multiplier of the SPX options is \$100 per index point and the average value of the index over the sample period is equal to 1,546. Hence, a unit position in the index represents an average investment of \$154,600. This scale may represent small retail investors. To capture portfolios of institutional investors, initial positions of 10, 100 and 1,000 times the index are also considered, which corresponds to portfolios of \$1,55m, \$15,5m and \$155m, respectively.

A larger scale naturally requires a larger transaction volume and thus involves stronger trade impact. To capture this effect, position limits are imposed based on the depth of the price quotes: $\alpha \leq s(\mathbf{a})$, $\beta \leq s(\mathbf{b})$, where $s(\mathbf{a})$ and $s(\mathbf{b})$ are the quoted best ask size and best bid size, respectively.

These constraints are permissive in the sense that the quoted volume may no longer be available when the buy or sell orders are placed. On the other hand, the constraints are restrictive in the sense that the volumes offered at less favorable prices are not considered, because the data set includes only the best bid and ask prices and Level-II data about the order book were not available for this study.

The market depth constraints differ in a subtle way from the single-option restriction of Constantinides, Czerwonko and Perrakis (2019, Section 4), which restricts the total option position to one option per one unit of index, regardless of portfolio scale and quoted volumes.

The effect of the market depth constraints depends on the portfolio size: they have no material effect for the unscaled portfolio but they become highly relevant for the scaled portfolios (10, 100 or 1,000 times the index).

A separate analysis is performed based on the single-option restriction instead of the market depth constraints. Using this specification, the optimal portfolio returns and market index returns are guaranteed to be co-monotonic, and the SA system becomes indistinguishable from the single-crossing rule.

Solvency constraints and margin constraints have a minimal effect on the analysis because the SSD constraints force the solution portfolio to have less downside risk than the index. Among other things, SSD requires that the solution portfolio has a higher minimum return than the index, which implies that the enhanced portfolio is solvent in every scenario: $\mathbf{X}'\boldsymbol{\alpha}^* - \mathbf{X}'\boldsymbol{\beta}^* \geq -y$.

To the extent that margin requirements are designed to ensure solvency of the writer, the SSD restrictions thus reduce the need for explicit margin requirements. By contrast, the dual alternative to the CJP system (which does not give a sufficient condition for SSD by the solution portfolio) does not ensure solvency and would benefit from adding explicit solvency constraints or margin requirements.

4.3 Statistical testing

Statistical tests can be used to determine whether the out-of-sample dominance relations between the enhanced portfolio and the index are statistically significant.

Regretfully, small-sample tests for SSD are difficult to obtain due to the model-free assumption framework and the complexity of the hypothesis structure. However, several large-sample tests are available.

Most statistical tests use a null hypothesis of dominance and an alternative of non-

dominance. This hypothesis structure seems well suited for the evaluation of out-of-sample returns of portfolios which are constructed to dominate the index.

The present study focuses on the Linton, Maasoumi and Whang (2005) test for SSD, which uses a bootstrap method to estimate critical values for a Kolmogorov-Smirnov type test statistic. In the (rare) case of a perfect out-of-sample dominance relation, the LMW05 test statistic takes a value of zero and the estimated p-value takes a value of one.

If dominance cannot be rejected, it becomes relevant to also test for non-dominance, to avoid false dominance classifications due to a possible lack of statistical power of the SSD test. For this purpose, the bootstrap Empirical Likelihood Ratio test by Davidson (2009) is used, with cutoffs at the first and ninth decile break points of the distribution of out-of-sample SPX returns. If empirical violations of SSD occur for return levels which lie between the two cutoffs, then the D09 test statistic takes a value of zero while the LMW05 test statistic takes a strictly positive value.

4.4 Results

Table III summarizes the portfolio characteristics and investment performance of the passive strategy and the enhanced portfolio for various scales. The focus is on using 28-day options and the entire sample period. Results for alternative specifications will be presented and discussed later on.

For the unscaled strategy ($S = 1$), SA opportunities which generate current income in excess of one index point (\$100) are found for each of the 168 formation dates. A passive investor in the index can thus improve her performance by adding an active options overlay. The optimal overlay takes a net short position in calls and a net long position puts, which suggests that downside risk is reduced by sacrificing upside potential. The combined option positions result in a negative delta and a positive vega for the enhanced portfolio.

Naturally, the optimization problem ignores possible specification error and parameter estimation error for the probability distribution of index returns (see Section 3.1). For this reason, the investment performance is reported both in sample and out of sample. Encouragingly, significant performance improvements are found out of sample. The average excess return of the passive portfolio is boosted by an impressive 19.03% per annum; a high positive skewness of 10.41 is achieved; the annualized Sortino ratio takes a value of 1.90.

The enhanced portfolio is designed to appeal to all risk averse investors. Indeed, improvements in the Certainty Equivalent Returns (CERs) are achieved for a wide range of risk preferences. An approximate out-of-sample dominance relationship is achieved, with limited empirical violations of SSD. The LMW05 test cannot reject dominance (p-value: 0.41). Since minor empirical violations of dominance occur, the D09 test takes a value of zero (p-value: 1.00).

The above results are consistent with those in the aforementioned earlier studies of active options trading strategies. Unfortunately, the results appear not robust to the assumed portfolio size. Many of the optimal option positions are equal to or approach the quoted bid or ask size for the relevant option series, which raises concerns about the scalability of the strategy. Indeed, the investment performance quickly deteriorates as the scale of the portfolio is increased. Scaling the portfolio reduces the economic and statistical significance of outperformance and eventually leads to underperformance.

Already for 10 times the index ($S = 10$), or a \$1,55m portfolio, the enhanced portfolio underperforms the index out of sample. The LMW05 test rejects dominance at conventional significance levels; the D09 test of course confirms that there exists no empirical evidence non-dominance.

Since the analysis ignores trading fees and slippage risk, the profitability of the large-scale strategies is likely to be even lower in practice than is estimated here.

[Insert Table III about here.]

Figure 1 and Figure 2 illustrate the typical portfolio composition of the options layover portfolio. Every panel ($S=1, 10, 100, 1000$) shows the quartile break points (across the formation dates) for the number of options at a given level of moneyness (k_i/y_0). For many moneyness levels, the median position is close to zero and the distribution is highly skewed to negative values (short positions) or highly skewed to positive values (long positions), so that only the first or third quartile breakpoint takes a non-zero value. For both calls and puts, long positions in at-the-money contracts and short positions in out-of-the-money contracts occur frequently but not simultaneously.

In both figures, the scale of the ordinate in the four panels is proportional to the portfolio scale (S). If the market depth constraints would not be binding, the patterns in the four panels would look identical. The differences between the panels therefore reflect the lack of scalability due to binding constraints.

[Insert Figure 1 about here.]

[Insert Figure 2 about here.]

Figure 3 further illustrates the results. Every panel demonstrates the typical conditional risk profile of the enhanced portfolio by plotting the quartile break points (across all expiration dates) of the investment return against the index return. The figure uses the feature that the returns to all option series are fully determined by the index return. The most noticeable pattern is a reduction of downside risk accompanied by a reduction of upside potential. This pattern reflects the writing of calls and buying of puts on a net basis. The differences between the patterns in the four panels again reflects the lack of scalability due to binding market depth constraints.

[Insert Figure 3 about here.]

Figure 4 illustrates the approximate out-of-sample dominance relation by comparing the out-of-sample empirical CDF of the enhanced portfolio with that of the SPX. The improvements of mean return, skewness and left tail risk are clearly visible and an approximate SSD relation emerges, for the unscaled strategy ($S = 1$). As the portfolio scale increases, the distribution of the enhanced portfolio however quickly deteriorates and significant violations of dominance occur.

[Insert Figure 4 about here.]

The results are surprisingly robust to the exclusion of the 2007-2009 period with extreme market volatility levels. The robustness is attributable to the use of market information to obtain conditional probability estimators. If unconditional estimators are used, then the results become very sensitive for the sample period.

During the Global Financial Crisis (GFC), market volatility and options prices climbed to unprecedented levels. Using an unconditional estimator based on a constant volatility level, option prices appear very expensive, providing an strong incentive for writing options. Since the unconditional estimator assigns minimal probability to double-digit losses, the optimal enhanced portfolio writes deep-out-of-the-money put options on a large scale. This strategy is very profitable but also extremely risky. Despite the very high dispersion, the strategy is quite fortunate during the GFC: it narrowly escapes large losses during the largest market down swings and generates a bumper profit in the month of December 2008 when market volatility unexpectedly fell from historical highs.

The analysis is further extended by changing the time to expiration of the traded options. Consistent with Constantinides, Czerwonko and Perrakis (2019), the profitability of the strategy falls when back-month options are considered. Table IV shows the results for three-month options. The investment performance deteriorates significantly compared with that based on the 28-day options in Table III by every performance measure. The SSD relation

is now only marginally significant for the unscaled portfolio, using the LMW05 test.

[Insert Table IV about here]

Table V shows the results for 7-day options, assuming that the portfolio is fully invested in the SPX when 7-day options are not available. The strategy is highly profitable over the life the option, but the overall monthly returns are lowered by not having an open options position for three fourths of the month. Weekly SPX options are not included in this study, in contrast to Constantinides, Czerwonko and Perrakis (2019). Using weeklies, investment in one-week options outside of the expiration week of the monthlies becomes possible. Loosely speaking, the outperformance of the index can be expected to be four times as high as based on monthly one-week options, assuming that the weeklies contain similar SA opportunities as the monthlies.

[Insert Table V about here]

Table VI summarizes the effect of various alternative specifications for the market risk premium (MRP) in addition to the baseline specification with $MRP = 0.04$. The alternatives are $MRP = 0$, $MRP = 0.08$, $MRP \propto VIX$, and $MRP \propto VIX^2$. The results appear highly robust to the specification, and suggest that the estimation of the market risk premium is less relevant than the estimation of market volatility, for trading options. This finding is reminiscent of the invariance of the theoretical value of options to the drift rate of the underlying asset in complete and perfect markets.

[Insert Table VI about here]

Table VII summarizes results which are obtained if the market depth constraints are replaced with the single-option restriction. Since the single-option restriction does not account for the available market depth, there is no explicit scale for the results based on this restriction. However, a maximum scale can be computed a posteriori based on the quoted volume of the option series which are included in the optimal portfolio. For every maturity, the number of formation dates is computed for which the optimal portfolio can be scaled by a factor of $S = 1, 10, 100, 1000$.

The single-option rule tends to be more restrictive than the market depth constraints and leads to performance deterioration, for $S = 1$. Furthermore, the options layover often takes concentrated positions which have limited scalability; a scale of $S = 100$ is generally not feasible and even $S = 10$ appears often problematic, for every maturity. More worryingly, the hypothesis of dominance is rejected at conventional significance levels for every maturity, which means that outperformance becomes elusive at every portfolio scale.

[Insert Table VII about here]

5 Conclusions

The CJP system (2) amounts to a set of first-order optimality conditions for Convex Optimization. These conditions are computationally efficient for testing the optimality of the passive stock index but they do not identify the dominating alternatives if the index is sub-optimal. By contrast, the larger SA system captures all feasible risk arbitrage opportunities and thus better captures the economic significance of pricing errors if the market is inefficient.

The empirical application shows that systematic SA opportunities exist for front-month SPX options, consistent with the conclusions of CJP, Constantinides, Czerwonko, Jackwerth

and Perrakis (2011) and Constantinides, Czerwonko and Perrakis (2019). However, the economic significance seems to depend critically on the assumed portfolio size. Ex-ante outperformance seems elusive for large-scale implementation of the proposed trading strategy, if market depth is taken into account. In fact, the large-scale strategy underperforms a passive position in the SPX out of sample, due to the bid-ask spreads.

A number of extensions are left for further research. The computed investment performance may worsen if slippage risk is taken into account: SA opportunities may disappear if competitors pick up bargains before the investor does. On the other hand, performance could improve if the investor uses execution timing algorithms to exploit high-frequency predictability of options prices. For algorithmic traders, the quoted bid-ask spread can overstate the effective costs of trading options, as demonstrated by Muravyev and Pearson (2016).

Results of preliminary computations however temper the expectations about large-scale SA opportunities inside the bid-ask range. The analysis for 28-day options and $S = 1000$ (a \$155 million investment in the index) was repeated under the assumption that trading is possible at the bid-ask midpoint, or $\mathbf{a}^* = \mathbf{b}^* = \frac{1}{2}(\mathbf{a} + \mathbf{b})$, so that the bid-ask spread is reduced to zero. Although this approach improves average return by several percentage points per annum, the active portfolio still trails the passive benchmark out of sample.

To further analyze the scalability of SA opportunities, Level II data about the order book could be used to assess the price impact of large transactions. This approach would account for the volumes offered at prices above the best ask prices and requested at the best bid prices. The large performance deterioration observed in this study for the relatively low scaling level of $S = 10$ (a \$1.55 million investment in the index.) however mitigates the expectations for finding substantially more 'money left on the table'.

The inclusion of weekly SPX options may uncover larger-scale SA opportunities for short maturities. Using only monthly options, the optimal portfolio for $S = 10$ adds 6.78% annually to the index return without significant violation of SSD (p-value: 0.22), in Table V. The inclusion of weeklies is expected to further enhance performance, by allowing open option po-

sition throughout the entire month, and, furthermore, facilitates larger-scale implementation by relaxing the market depth constraints.

The strong divergence between results in sample and out of sample suggests that there is scope for further improvement of the probability estimation. The Cox, Ross and Rubinstein (1979) model could be replaced by a more sophisticated lattice model such as Rubinstein's (1998) Edgeworth binomial tree which incorporates skewness and kurtosis of the index.

Implied probabilities are another interesting route for further research. Notably, the Smoothed Empirical Likelihood method by Kitamura, Tripathi and Ahn (2004) can provide a consistent, non-parametric and discrete estimator for the conditional density. The Peso problem associated with implied probabilities could be mitigated by using a sufficiently long estimation window which includes the bear markets of the 1930s and 1970s.

The definition of SA in this study is based on the SSD relation. Additional risk arbitrage opportunities may be uncovered using third-degree stochastic dominance (TSD) which assumes decreasing risk aversion in addition to global risk aversion. Ritchken and Kuo (1989) demonstrate that increasing the degree of SD leads to tighter pricing bounds for individual options. Similarly, increasing the degree of SD amounts to relaxing the SA system and enlarging the set of dominating portfolios. Post and Kopa (2017) develop Quadratic Programming solutions for portfolio optimization with TSD constraints.

Other possible relaxations of the SA system include the use of Almost Stochastic Dominance (Leshno and Levy (2002)) and the weakening of dominance constraints in empirical 'contact areas' (Linton, Post and Whang (2014)). Relaxing instead of tightening the SA system seems in order because sampling variation tends to fuel false non-dominance classifications rather than false dominance classifications (Kroll and Levy (1980)).

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Appendix

Proof to Theorem 2.3.3: The strategy is to first derive the dual alternative to system (2) and next to show that the alternative has a solution if and only if the system (3) is solvable.

To allow for compact notation, let $\mathbf{P} := (\mathbf{1}_M \mathbf{p})$ and $\mathbf{C} := (\mathbf{P} \circ \mathbf{X}) \mathbf{U}$, where $\mathbf{U}$ is an upper triangular summation matrix. Let $\boldsymbol{\gamma} = \mathbf{D}\boldsymbol{\mu}$ be the decrements of the kernel, so that $\mathbf{U}\boldsymbol{\gamma} = \boldsymbol{\mu}$. System (2) can be reduced to

$$\begin{aligned} \mathbf{C}\boldsymbol{\gamma} &\leq \mathbf{a}; \\ -\mathbf{C}\boldsymbol{\gamma} &\leq -\mathbf{b}; \\ \boldsymbol{\gamma} &\geq \mathbf{0}_N. \end{aligned} \tag{6}$$

By Farkas' Lemma (Gale, Kuhn and Tucker (1951, Lemma 1, p. 318)), a solution to the following alternative exists if and only if system (6) is not solvable:

$$\begin{aligned} \mathbf{C}'\boldsymbol{\alpha} - \mathbf{C}'\boldsymbol{\beta} &\geq \mathbf{0}_N; \\ \mathbf{a}'\boldsymbol{\alpha} - \mathbf{b}'\boldsymbol{\beta} &< 0; \\ \boldsymbol{\alpha}, \boldsymbol{\beta} &\geq \mathbf{0}_M. \end{aligned} \tag{7}$$

This system requires the existence of an overlay portfolio which features negative net costs $(\mathbf{a}'\boldsymbol{\alpha} - \mathbf{b}'\boldsymbol{\beta})$ and non-negative partial co-moments $\mathbf{C}'_j(\boldsymbol{\alpha} - \boldsymbol{\beta}) = C(\mathbf{x}'(\boldsymbol{\alpha} - \boldsymbol{\beta})|y_j)$, $j = 1, \dots, N$, where $C(x|t) := \mathbb{E}_{\mathcal{F}}[x\mathcal{I}(y \leq t)]$.

Every solution $(\boldsymbol{\alpha}, \boldsymbol{\beta})$ to system (3) is also a solution to system (7), regardless of the specification of $\mathcal{P}$. To see this, let $x := \mathbf{x}'\boldsymbol{\alpha} - \mathbf{x}'\boldsymbol{\beta}$ and rewrite the lower partial moment inequality $L(x + y|y_j) \leq L(y|y_j)$, $j = 1, \dots, N$, as

$$L(y|y_j) - C(x|y_j) + R(x, y|y_j) \leq L(y|y_j), \quad (8)$$

where $R(x, y|y_j) := \mathbb{E}_{\mathcal{F}}[(y_j - x - y)(\mathcal{I}(x + y \leq y_j) - \mathcal{I}(y \leq y_j))]$. Due to co-monotonicity of $(y_j - x - y)$ and $(\mathcal{I}(x + y \leq y_j) - \mathcal{I}(y \leq y_j))$, $R(x, y|y_j) \geq 0$, and (8) implies the co-moment inequality $C(x|y_j) \geq 0$, which suffices to prove that that system (7) is solvable if system (3) is solvable.

Furthermore, every solution (α, β) to system (7) implies the existence of a solution $(\epsilon\alpha, \epsilon\beta)$, $\epsilon > 0$, to system (3). Let $x_j := \mathbf{x}'_j\alpha - \mathbf{x}'_j\beta$, $j = 1, \dots, N$. Since $y_1 < \dots < y_N$, there exists $\epsilon > 0$ which is sufficiently small to ensure that $\epsilon x_j + y_j < y_k$ and $y_j < \epsilon x_k + y_k$ for all $j = 1, \dots, N-1$, and $k > j$. For this small value of $\epsilon > 0$, $\mathcal{I}(\epsilon x_k + y_k \leq y_j) = \mathcal{I}(y_k \leq y_j)$ for all $k \neq j$. Consequently, the following equality applies for every $j = 1, \dots, N$:

$$L(\epsilon x + y|y_j) = L(y|y_j) - S(\epsilon x|y_j), \quad (9)$$

where $S(x|y_1) := \min(0, C(x|y_1))$ and $S(x|y_j) := \min(C(x|y_{j-1}), C(x|y_j))$, $j = 2, \dots, N$. The co-moment inequalities $C(x|y_j) \geq 0$, $j = 1, \dots, N$, imply $S(x|y_j) \geq 0 \Leftrightarrow S(\epsilon x|y_j) \geq 0$, $j = 1, \dots, N$, and therefore (via (9)) the lower partial moment inequalities $L(\epsilon x + y|y_j) \leq L(y|y_j)$, for sufficiently small $\epsilon > 0$. In addition, feasibility, or $(\epsilon\alpha, \epsilon\beta) \in \mathcal{P}$, can be established for sufficiently small $\epsilon > 0$, because the portfolio constraints are not binding at $(\alpha, \beta) = (\mathbf{0}_M, \mathbf{0}_M)$. ■

Proof of Corollary 2.3.4: Let $D_1(z, y|t) := F_z(t) - F_y(t)$ and $D_2(z, y|t) := L(z|t) - L(y|t)$. Since $F_x(t) = \partial L(x|t)/\partial t$, it follows that $D_1(z, y|t) = \partial D_2(z, y|t)/\partial t$. The stated conditions imply that $D_2(z, y|y_1) = 0$ (by $z \succeq_{SSD} y$ and $L(y|y_1) = 0$), $D_2(z, y|y_2) < 0$ and $D_2(z, y|y_N) < 0$. Therefore, if $D_2(z, y|y_j) = 0$, for some $j = 3, \dots, N-1$, then it follows that $D_2(z, y|t)$ has at least one local maximum and at least one local minimum, and, furthermore, the derivative $D_1(z, y|t)$ changes sign at least twice. Since $D_1(z, y|t) = F_z(t) - F_y(t)$, this situation requires at least two CDF crossings. The condition $L(z|y_N) < L(y|y_N)$, or $D_2(z, y|y_N) < 0$, avoids the situation in which $D_2(z, y|t) = 0$ for all thresholds after the first local minimum, which would amount to a contact point instead of a crossing point. ■

Proof of Proposition 3.2.1: The SSD restriction $\mathbf{x}'\alpha - \mathbf{x}'\beta + y \succeq_{SSD} y$ amounts to $L(\mathbf{x}'\alpha - \mathbf{x}'\beta + y|y_j) \leq L(y|y_j)$ for all y_j , $k = 1, \dots, N$. Using $L(x + y|y_j) = \mathbb{E}_{\mathcal{F}}(y_j - x - y)_+$, where $(\cdot)_+$ is the element-wise positive part operator, the N lower partial moment constraints can be stacked as follows:

$$\mathbf{p}(\mathbf{1}_T \mathbf{y} - \mathbf{X}'\alpha \mathbf{1}'_N + \mathbf{X}'\beta \mathbf{1}'_N - \mathbf{y}'\mathbf{1}'_N)_+ \leq \mathbf{L}. \quad (10)$$

Using a linearization based on Rockafellar and Uryasev (2000) and Dentcheva and Ruszczyński (2003), this system can be rewritten as

$$\begin{aligned}
p\Theta &\leq L; \\
-\Theta - \mathbf{X}'\alpha\mathbf{1}'_N + \mathbf{X}'\beta\mathbf{1}'_N &\leq \mathbf{y}'\mathbf{1}'_N - \mathbf{1}_N\mathbf{y}; \\
\Theta &\geq \mathbf{0}_{N \times N}.
\end{aligned} \tag{11}$$

The additional model variables Θ capture the element-wise positive parts; the solution $\Theta = (\mathbf{1}_T\mathbf{y} - \mathbf{X}'\alpha\mathbf{1}'_N + \mathbf{X}'\beta\mathbf{1}'_N - \mathbf{y}'\mathbf{1}'_N)_+$ is feasible if (10) is satisfied. Replacing (10) with (11) in (3) gives (4). ■

Table I: Cross-sectional dimension.

Shown is the median number of different option series which are available at the formation date. Distinction is drawn between call and put, as well as bid and ask. Three time periods are distinguished which roughly correspond to before, during and after the Global Financial Crisis (GFC) of 2007-2009. Three different option maturities are considered: 91 days, 28 days and 7 days. Three different levels of moneyness are analyzed: Low, or $k_i/y_0 < 0.90$, refers to in-the-money calls and out-of-the-money puts; Medium, or $0.90 \leq k_i/y_0 \leq 1.05$, refers to at-the-money calls and puts; High, or $k_i/y_0 > 1.05$, refers to out-of-the-money calls and in-the-money puts.

Period	Maturity	Moneyness	Call		Put	
			Bid	Ask	Bid	Ask
2004-2006	7 days	Low	19	19	3	18
		Medium	33	35	34	35
		High	1	11	11	11
	28 days	Low	16	16	11	16
		Medium	33	33	33	33
		High	8	12	12	12
	91 days	Low	15	15	12	15
		Medium	16	16	16	16
		High	5	9	9	9
2007-2009	7 days	Low	35	35	17	35
		Medium	39	40	40	40
		High	17	51	51	51
	28 days	Low	38	38	28	38
		Medium	41	41	41	41
		High	29	49	49	49
	91 days	Low	15	15	15	15
		Medium	27	27	27	27
		High	27	35	35	35
2010-2018	7 days	Low	122	122	25	118
		Medium	58	59	59	59
		High	8	33	33	33
	28 days	Low	120	120	81	118
		Medium	60	60	60	60
		High	20	33	33	33
	91 days	Low	81	81	73	81
		Medium	40	40	40	40
		High	18	26	26	26

Table II: Distribution of quote size.

Shown is the median quote size per option series available at the formation date. Distinction is drawn between call and put, as well as bid and ask. Three time periods are distinguished which roughly correspond to before, during and after the Global Financial Crisis (GFC) of 2007-2009. Three different option maturities are considered: 91 days, 28 days and 7 days. Three different levels of moneyness are analyzed: Low, or $k_i/y_0 < 0.90$, refers to in-the-money calls and out-of-the-money puts; Medium, or $0.90 \leq k_i/y_0 \leq 1.05$, refers to at-the-money calls and puts; High, or $k_i/y_0 > 1.05$, refers to out-of-the-money calls and in-the-money puts.

Period	Maturity	Moneyness	Call		Put	
			Bid	Ask	Bid	Ask
2004-2006	7 days	Low	50	50	20	10
		Medium	50	50	50	50
		High	41	23	50	50
	28 days	Low	50	50	20	15
		Medium	50	50	50	50
		High	10	50	50	50
	91 days	Low	50	50	53	50
		Medium	53	53	53	53
		High	50	50	50	50
2007-2009	7 days	Low	55	55	35	54
		Medium	55	54	54	54
		High	22	53	55	101
	28 days	Low	54	54	50	51
		Medium	54	54	54	53
		High	51	60	101	62
	91 days	Low	62	101	101	100
		Medium	82	101	101	82
		High	100	101	101	62
2010-2018	7 days	Low	100	87	67	210
		Medium	82	75	101	89
		High	67	281	103	103
	28 days	Low	103	103	77	100
		Medium	94	100	162	153
		High	35	90	103	103
	91 days	Low	100	96	150	115
		Medium	114	112	175	161
		High	131	113	101	103

Table III: Characteristics and performance of active portfolios with one-month options

This table summarizes the results for 28-day options. A passive strategy is fully invested in the SPX. Active strategies are based on the market-depth constraints for scale $S = 1, 10, 100, 1000$. The first row shows the number of formation dates (out of a total of 168) at which an SA opportunity is detected which generates current income in excess of S index points ($\$100S$) in sample. Next, medians (across the 168 active portfolios) are shown for the contract value of calls bought, calls written, puts bought and puts written, and scaled by $100S$ dollars, as well as medians for selected Greeks: delta, gamma and vega, computed at the at-the-money implied volatility, and scaled by S . Several summary statistics are included for (annualized) investment performance in sample and out of sample: mean, standard deviation, skewness, Sortino ratio and Certainty Equivalent Returns (CERs) for a power utility function with Relative Risk Aversion coefficient of 2 and 5. The two bottom rows display p-values of two statistical tests: Linton, Maasoumi and Whang (2005) and Davidson (2009).

		SPX	Enhanced Portfolio			
			$S = 1$	$S = 10$	$S = 100$	$S = 1000$
SA Opps (#)			168	168	166	161
Contracts (avg $\$/100S$)	Calls Bought		30.68	21.09	20.65	46.41
	Calls Written		100.10	90.01	81.09	92.44
	Puts Bought		140.15	88.99	66.83	73.81
	Puts Written		112.30	26.39	17.12	15.91
Greeks (avg/ S)	Delta		-0.16	-0.16	-0.16	0.05
	Gamma		0.02	0.02	0.02	0.01
	Vega		381.16	378.61	389.46	329.08
In-sample excess returns	Mean	5.72	59.66	30.33	26.44	19.95
	St.Dev	17.48	16.85	16.66	16.89	15.42
	Skew	0.14	0.71	0.36	0.51	1.12
	Sortino	0.54	>100	>100	4.26	3.89
	CER-2	3.95	58.34	28.98	25.03	19.01
	CER-5	-0.94	54.13	24.72	20.73	15.46
Out-of-sample excess returns	Mean	9.24	28.27	-0.45	-5.60	-5.20
	St.Dev	15.79	61.97	18.44	17.27	13.76
	Skew	-1.56	10.41	0.67	0.16	-0.89
	Sortino	0.44	1.90	-0.03	-0.36	-0.38
	CER-2	8.08	14.32	-2.29	-7.11	-5.69
	CER-5	3.27	4.57	-7.41	-11.87	-9.09
SSD tests	LMW05		0.41	0.02	0.00	0.01
	D09		1.00	1.00	1.00	1.00

Table IV: Characteristics and performance of active portfolios with three-month options

This table summarizes the results for 91-day options. A passive strategy is fully invested in the SPX. Active strategies are based on the market-depth constraints for scale $S = 1, 10, 100, 1000$. The first row shows the number of formation dates (out of a total of 139) at which an SA opportunity is detected which generates current income in excess of S index points ($\$100S$) in sample. Next, medians (across the 139 active portfolios) are shown for the contract value of calls bought, calls written, puts bought and puts written, and scaled by $100S$ dollars, as well as medians for selected Greeks: delta, gamma and vega, computed at the at-the-money implied volatility, and scaled by S . Several summary statistics are included for (annualized) investment performance in sample and out of sample: mean, standard deviation, skewness, Sortino ratio and Certainty Equivalent Returns (CERs) for a power utility function with Relative Risk Aversion coefficient of 2 and 5. The two bottom rows display p-values of two statistical tests: Linton, Maasoumi and Whang (2005) and Davidson (2009).

		SPX	Enhanced Portfolio			
			$S = 1$	$S = 10$	$S = 100$	$S = 1000$
SA Opps (#)			139	139	138	130
Contract value (avg $\$/100S$)	Calls Bought		125.83	54.92	49.53	132.32
	Calls Written		309.37	247.96	208.46	170.76
	Puts Bought		170.44	138.40	111.61	161.82
	Puts Written		89.19	48.35	46.44	56.69
Greeks (avg/ S)	Delta		-0.01	0.01	0.03	0.12
	Gamma		0.00	0.00	0.00	0.00
	Vega		450.85	448.49	473.88	457.44
In-sample excess returns	Mean	5.64	20.07	16.58	16.09	13.55
	St.Dev	17.72	17.45	17.64	17.90	17.00
	Skew	0.26	0.54	0.61	0.81	1.10
	Sortino	0.57	>100	2.55	2.47	2.40
	CER-2	3.50	18.24	14.68	14.13	11.95
	CER-5	-1.38	13.93	10.31	9.76	8.09
Out-of-sample excess returns	Mean	8.76	-0.86	-4.10	-4.05	-4.90
	St.Dev	15.98	18.43	16.46	16.56	14.61
	Skew	-1.69	0.04	-0.27	-0.12	-0.80
	Sortino	0.41	-0.05	-0.25	-0.25	-0.32
	CER-2	7.02	-3.20	-5.82	-5.79	-6.11
	CER-5	0.59	-9.18	-10.91	-10.87	-10.64
SSD tests	LMW05		0.00	0.00	0.00	0.00
	D09		1.00	1.00	1.00	1.00

Table V: Characteristics and performance of active portfolios with monthly one-week options.

This table summarizes the results for monthly 7-day options. A passive strategy is fully invested in the SPX. Active strategies hold the SPX for 28 days and enhance this position with the optimal portfolio of options during the last week of this period. Active strategies are based on the market-depth constraints for scale $S = 1, 10, 100, 1000$. The first row shows the number of formation dates (out of a total of 168) at which an SA opportunity is detected which generates current income in excess of S index points ($\$100S$) in sample. Next, medians (across the 168 active portfolios) are shown for the contract value of calls bought, calls written, puts bought and puts written, and scaled by $100S$ dollars, as well as medians for selected Greeks: delta, gamma and vega, computed at the at-the-money implied volatility, and scaled by S . Several summary statistics are included for (annualized) investment performance in sample and out of sample: mean, standard deviation, skewness, Sortino ratio and Certainty Equivalent Returns (CERs) for a power utility function with Relative Risk Aversion coefficient of 2 and 5. The two bottom rows display p-values of two statistical tests: Linton, Maasoumi and Whang (2005) and Davidson (2009).

		SPX	Enhanced Portfolio			
			$S = 1$	$S = 10$	$S = 100$	$S = 1000$
SA Opps (#)			168	168	166	161
Contract value (avg $\$/100S$)	Calls Bought	30.68	21.09	20.65	46.41	
	Calls Written	100.10	90.01	81.09	92.44	
	Puts Bought	140.15	88.99	66.83	73.81	
	Puts Written	112.30	26.39	17.12	15.91	
Greeks (avg/ S)	Delta	-0.16	-0.16	-0.16	0.05	
	Gamma	0.02	0.02	0.02	0.01	
	Vega	381.16	378.61	389.46	329.08	
In-sample excess returns	Mean	5.76	108.29	53.24	43.23	23.26
	St.Dev	17.96	16.95	16.56	16.28	13.52
	Skew	0.07	0.64	0.32	0.48	0.91
	Sortino	0.50	>100	9.78	7.62	5.45
	CER-2	3.80	106.85	51.88	41.95	22.82
	CER-5	-1.34	102.57	47.64	37.87	19.99
Out-of-sample excess returns	Mean	9.24	21.62	6.78	3.52	2.85
	St.Dev	15.79	24.78	14.13	14.13	14.08
	Skew	-1.56	5.36	-0.52	-1.04	-1.39
	Sortino	0.44	1.64	0.45	0.23	0.17
	CER-2	8.08	18.49	6.24	2.94	2.26
	CER-5	3.27	13.36	2.79	-0.66	-1.47
SSD tests	LMW05	0.87	0.22	0.06	0.00	
	D09	0.25	1.00	1.00	1.00	

Table VI: The effect of the assumed specification of the market risk premium

This table summarizes results for monthly 7-day, 28-day and 91-day options for various alternative specifications for the market risk premium. Active strategies are based on the market-depth constraints for scale $S = 1$. The results for the baseline specification $MRP = 0.04$ are taken from Tables III, IV and V; specifications $MRP = 0$ and $MRP = 0.08$ change the general level of the risk premium; specifications $MRP \propto VIX$ and $MRP \propto VIX^2$ assume that the risk premium is proportional to the level or square of concurrent value of the VIX index. Several summary statistics are included for (annualized) investment performance out of sample: mean, standard deviation, skewness, Sortino ratio and Certainty Equivalent Returns (CERs) for a power utility function with Relative Risk Aversion coefficient of 2 and 5. The two bottom rows display p-values of two statistical tests: Linton, Maasoumi and Whang (2005) and Davidson (2009).

			Market risk premium				
			0.04	0	0.08	$\propto VIX$	$\propto VIX^2$
28-day options	Out-of-sample excess returns	Mean	28.27	27.36	29.51	28.55	28.46
		St.Dev	61.97	61.96	61.78	61.94	61.89
		Skew	10.41	10.41	10.42	10.41	10.38
		Sortino	1.90	1.87	1.99	1.95	1.92
		CER-2	14.32	13.42	15.64	14.60	14.50
		CER-5	4.57	3.79	5.86	4.85	4.69
	SSD tests	LMW05	0.41	0.34	0.41	0.43	0.40
		D09	1.00	1.00	1.00	1.00	1.00
91-day options	Out-of-sample excess returns	Mean	-0.86	-1.57	0.57	-0.75	-0.66
		St.Dev	18.43	18.34	18.19	18.47	18.77
		Skew	0.04	0.23	-0.15	0.04	0.11
		Sortino	-0.05	-0.10	0.03	-0.05	-0.04
		CER-2	-3.20	-3.82	-1.74	-3.11	-3.12
		CER-5	-9.18	-9.37	-7.96	-9.13	-9.27
	SSD tests	LMW05	0.00	0.00	0.00	0.00	0.00
		D09	1.00	1.00	1.00	1.00	1.00
7-day options	Out-of-sample excess returns	Mean	21.62	21.63	21.78	21.64	20.95
		St.Dev	24.78	24.86	24.73	24.77	24.68
		Skew	5.36	5.33	5.35	5.39	5.51
		Sortino	1.64	1.66	1.56	1.65	1.61
		CER-2	18.49	18.47	18.66	18.52	17.89
		CER-5	13.36	13.30	13.49	13.41	12.86
	SSD tests	LMW05	0.87	0.88	0.86	0.86	0.87
		D09	0.25	0.24	0.19	0.21	0.23

Table VII: The effect of the single-option restriction

This table compares results based on market depth constraints with those based on the single-option restriction. A passive strategy is fully invested in the SPX. Active strategies use monthly 7-day, 28-day or 91-day options. Results based on the market depth constants with scale $S = 1, 10, 100, 1000$ are taken from Tables III, IV and V. Several summary statistics are included for (annualized) investment performance out of sample: mean, standard deviation, skewness, Sortino ratio and Certainty Equivalent Returns (CERs) for a power utility function with Relative Risk Aversion coefficient of 2 and 5. The two bottom rows display p-values of two statistical tests: Linton, Maasoumi and Whang (2005) and Davidson (2009). The bottom line of every panel shows which percentage of the optimal portfolios with single-option constraints can be scaled by a factor of $S = 1, 10, 100, 1000$, given the quote sizes.

			SPX	Single option	Market depth constraints			
					$S = 1$	$S = 10$	$S = 100$	$S = 1000$
28-day options	Out-of-sample excess returns	Mean	9.24	6.36	28.27	-0.45	-5.60	-5.20
		St.Dev	15.79	11.98	61.97	18.44	17.27	13.76
		Skew	-1.56	-2.92	10.41	0.67	0.16	-0.89
		Sortino	0.44	0.45	1.90	-0.03	-0.36	-0.38
		CER-2	8.08	6.28	14.32	-2.29	-7.11	-5.69
		CER-5	3.27	3.25	4.57	-7.41	-11.87	-9.09
	SSD tests	LMW05		0.07	0.41	0.02	0.00	0.01
		D09		1.00	1.00	1.00	1.00	1.00
	Scalability (%)				100	66.07	40.54	0
91-day options	Out-of-sample excess returns	Mean	8.76	6.53	-0.86	-4.10	-4.05	-4.90
		St.Dev	15.98	13.05	18.43	16.46	16.56	14.61
		Skew	-1.69	-2.73	0.04	-0.27	-0.12	-0.80
		Sortino	0.41	0.33	-0.05	-0.25	-0.25	-0.32
		CER-2	7.02	5.66	-3.20	-5.82	-5.79	-6.11
		CER-5	0.59	0.73	-9.18	-10.91	-10.87	-10.64
	SSD tests	LMW05		0.00	0.00	0.00	0.00	0.00
		D09		1.00	1.00	1.00	1.00	1.00
	Scalability (%)				100	85.61	64.71	0
7-day options	Out-of-sample excess returns	Mean	9.24	7.71	21.62	6.78	3.52	2.85
		St.Dev	15.79	14.30	24.78	14.13	14.13	14.08
		Skew	-1.56	-1.55	5.36	-0.52	-1.04	-1.39
		Sortino	0.44	0.42	1.64	0.45	0.23	0.17
		CER-2	8.08	7.04	18.49	6.24	2.94	2.26
		CER-5	3.27	3.15	13.36	2.79	-0.66	-1.47
	SSD tests	LMW05		0.10	0.87	0.22	0.06	0.00
		D09		1.00	0.25	1.00	1.00	1.00
	Scalability (%)				100	74.10	38.84	0

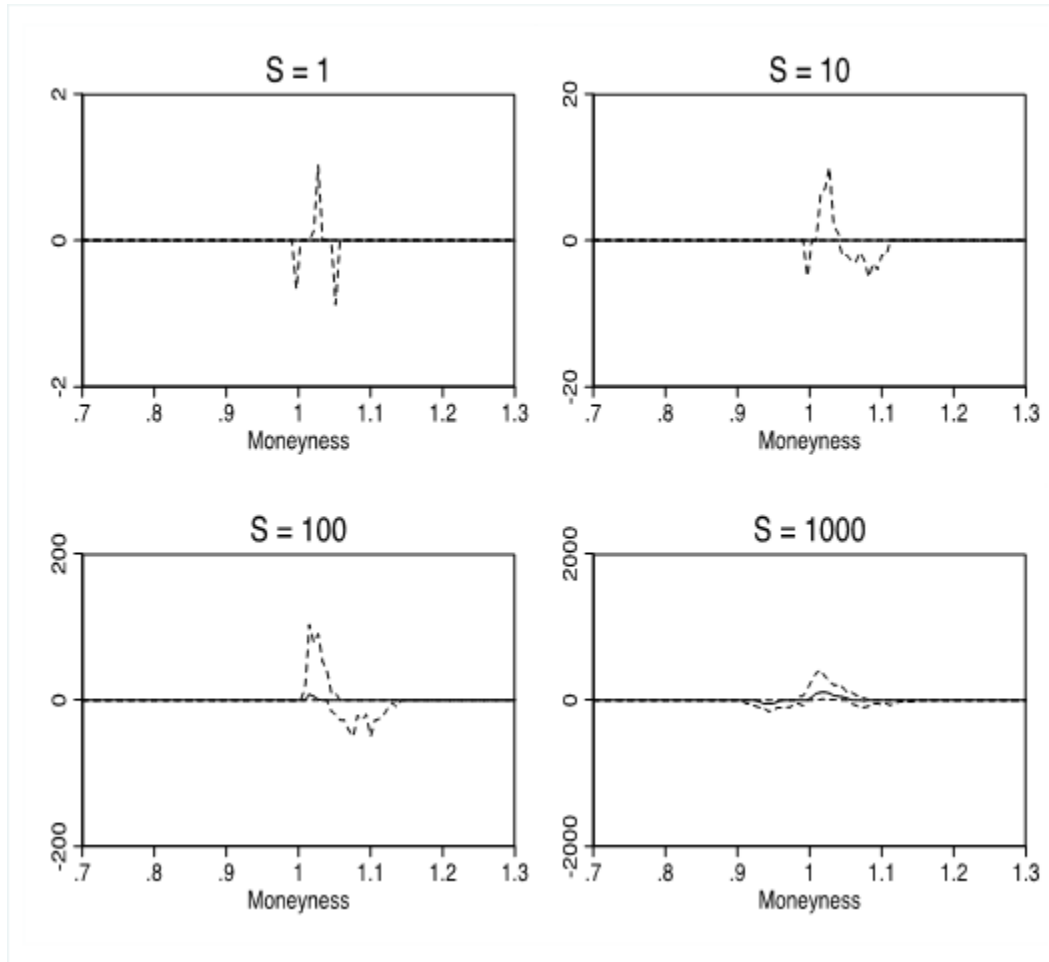


Figure 1: Optimal number of calls. Shown are the quartile break points (across all formation dates) for the number of call options in the optimal overlay portfolio at a given level of moneyness (k_i/y_0), for the analysis of 28-day SPX options. Each panel presents one of the four scaling levels: $S = 1, 10, 100, 1000$. The solid line represents the median; the dashed lines are the 25th and 75th percentiles.

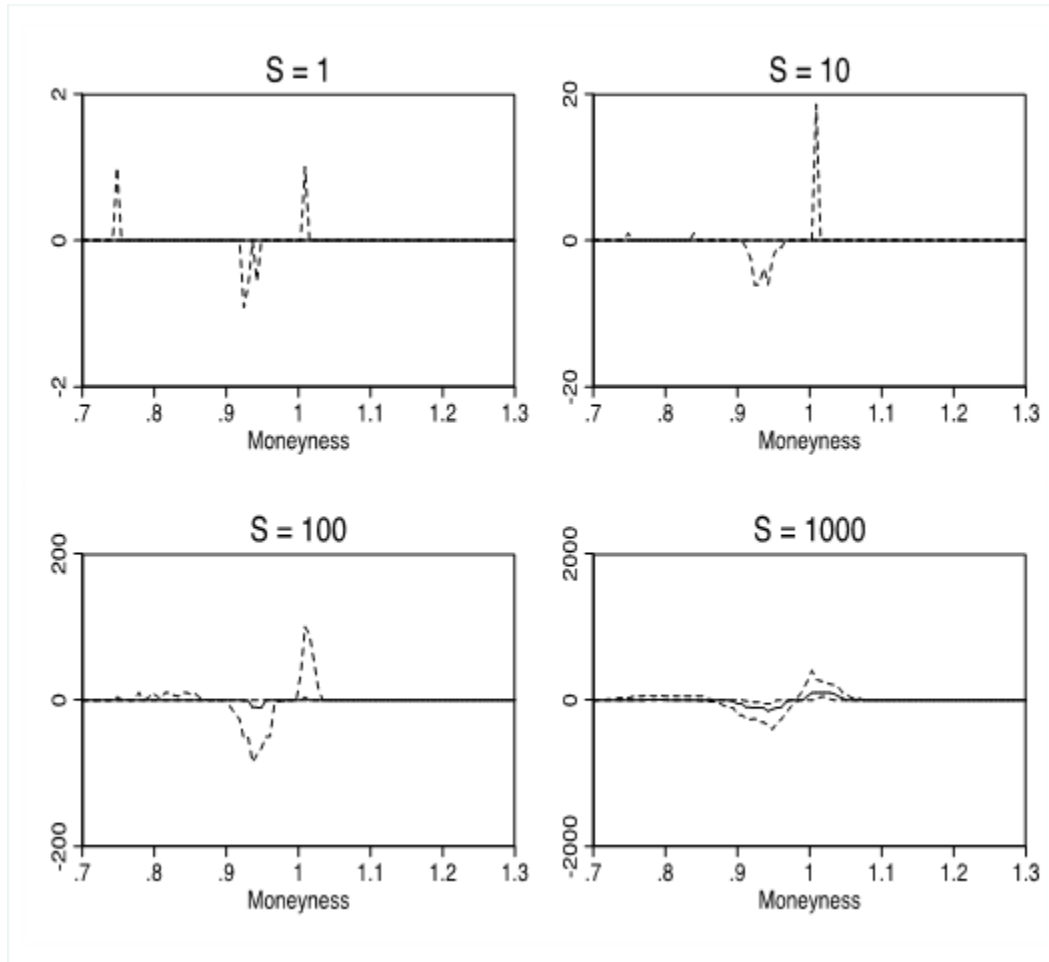


Figure 2: Optimal number of puts. Shown are the quartile break points (across all formation dates) for the number of put options in the optimal overlay portfolio at a given level of moneyness (k_i/y_0), for the analysis of 28-day SPX options. Each panel presents one of the four scaling levels: $S = 1, 10, 100, 1000$. The solid line represents the median; the dashed lines are the 25th and 75th percentiles.

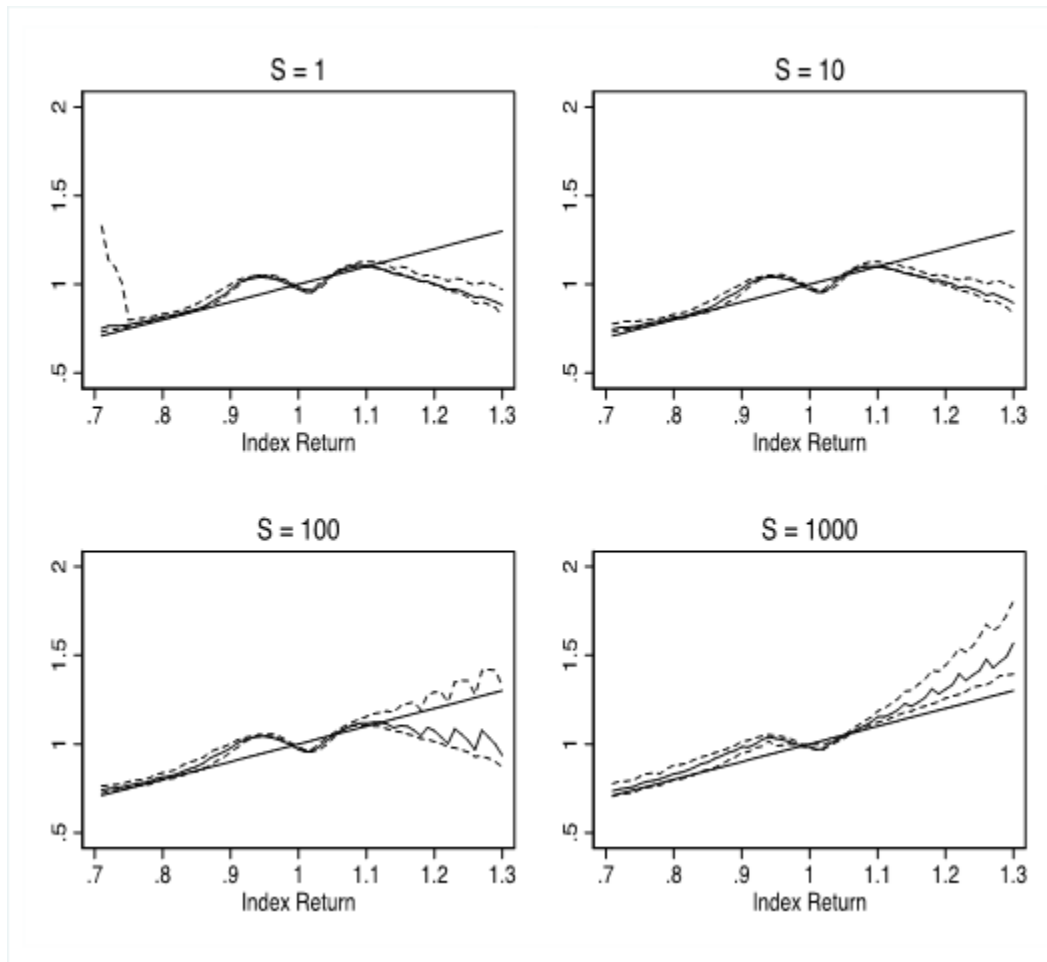


Figure 3: Conditional risk profile of the enhanced portfolio. This figure displays three quartile break points for ex-ante returns of the enhanced portfolios with 28-day options. Each graph presents one of the four scaling levels: $S = 1, 10, 100, 1000$. The solid line represents the median (across all expiration dates) of the active return conditional on the SPX; the dashed lines correspond to the first and third quartile break point, respectively. The bisector represents passive investment in the SPX.

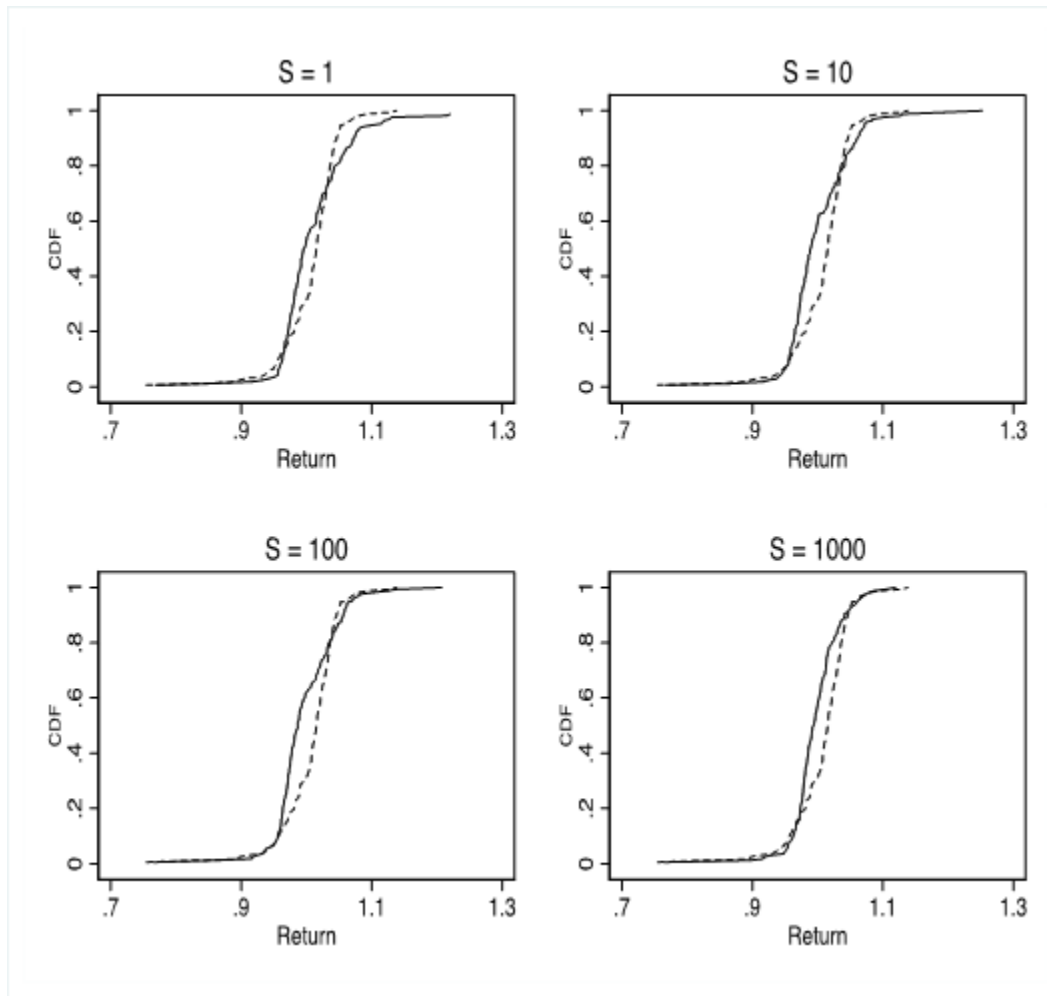


Figure 4: Empirical cumulative distribution functions of out-of-sample investment returns. This figure displays the empirical CDFs of the out-of-sample returns of the enhanced portfolios with 28-day options (solid lines) together with those of the corresponding passive strategy (dashed lines). Each panel represents one of the four scaling levels: $S = 1, 10, 100, 1000$.